Ethical AI: Fairness and Explainability in Income Prediction

(Adult Income Dataset – Logistic Regression with Fairlearn, SHAP, and LIME)

1. Introduction to Ethical AI

Ethical Artificial Intelligence (AI) focuses on building machine learning systems that are not only accurate but also fair, transparent, and accountable. As AI models are increasingly used in real-world decision-making—such as hiring, loan approvals, and income prediction—it is critical to ensure they do not reinforce or amplify societal biases.

Fairness in machine learning refers to the idea that model predictions should not systematically disadvantage individuals based on sensitive attributes such as gender, age, or race. Without fairness analysis, models may produce biased outcomes even when they achieve high accuracy.

Explainability is equally important, as complex models often behave like “black boxes.” Techniques such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) help interpret model decisions globally and locally, increasing transparency and trust.

This project evaluates fairness and explainability using a logistic regression model trained on the Adult Income dataset.

2. Dataset Description and Preprocessing

The dataset used in this study is the **Adult Income dataset**, which contains demographic and employment-related attributes used to predict whether an individual earns more than $50K per year.

Key Features:

- Age
- Education level
- Occupation
- Work hours per week
- Marital status
- Gender (sensitive attribute)

Target Variable:

- Income: <=50K or >50K (converted into binary classification)

Sensitive Attribute:

- Gender (sex) was selected for fairness evaluation.

Preprocessing Steps:

1. Removed irrelevant columns for prediction.
2. Converted target variable into binary format:
    - 1 = income >50K
    - 0 = income ≤50K
3. One-hot encoded categorical variables using pandas.get_dummies.
4. Split dataset into training (70%) and testing (30%) sets.
5. Preserved the sensitive attribute separately for fairness evaluation.

These preprocessing steps ensured that the dataset was suitable for both predictive modeling and fairness analysis.

3. Model Training and Evaluation

A **Logistic Regression model** was selected due to its interpretability and suitability for binary classification tasks.

Training Process:

- Model trained on preprocessed training dataset.
- Optimized using maximum iterations = 1000 to ensure convergence.

Evaluation Metrics:

The model was evaluated using:

- Accuracy score
- Confusion matrix
- Precision, recall, and F1-score (classification report)

Key Results:

The model achieved strong predictive performance, indicating that demographic and employment features are effective for income prediction. However, accuracy alone is not sufficient to evaluate real-world performance, especially when fairness concerns are present.

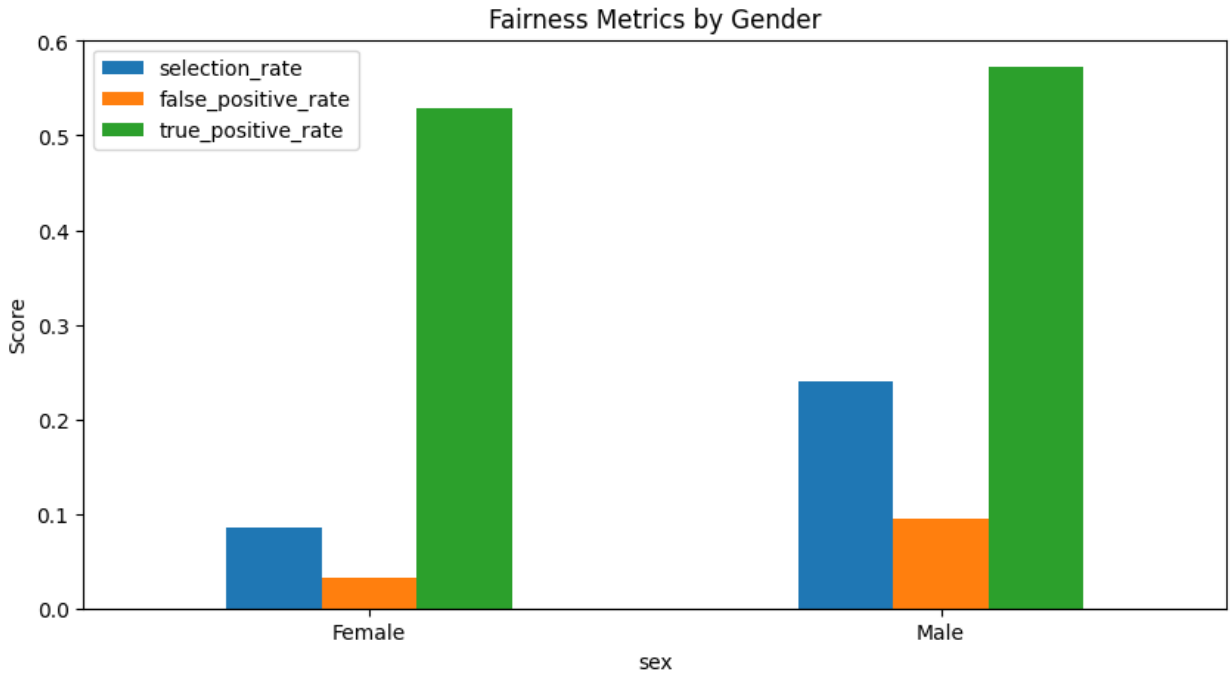
4. Fairness Analysis

Fairness was evaluated using the **Fairlearn** library. The analysis focused on differences in model behavior across gender groups.

Metrics Used:

- Selection Rate (positive prediction rate)

- False Positive Rate (FPR)
- True Positive Rate (TPR)



Findings:

The MetricFrame analysis revealed disparities between male and female groups:

- Selection rates differed across genders, indicating unequal likelihood of being predicted as high income.
- False Positive Rates varied, showing that one group experienced more incorrect positive predictions.
- True Positive Rates were not equal, indicating differences in model sensitivity across groups.

Interpretation:

These results suggest that the model exhibits **group-level bias**, even though it was trained without explicitly using gender as a feature in prediction. This may occur due to correlations between gender and other features (e.g., occupation, education).

Visualization:

Bar plots of fairness metrics clearly showed differences across gender groups, making bias easier to interpret and communicate.

5. Explainability Analysis

To ensure transparency, two explainability methods were applied: SHAP and LIME.

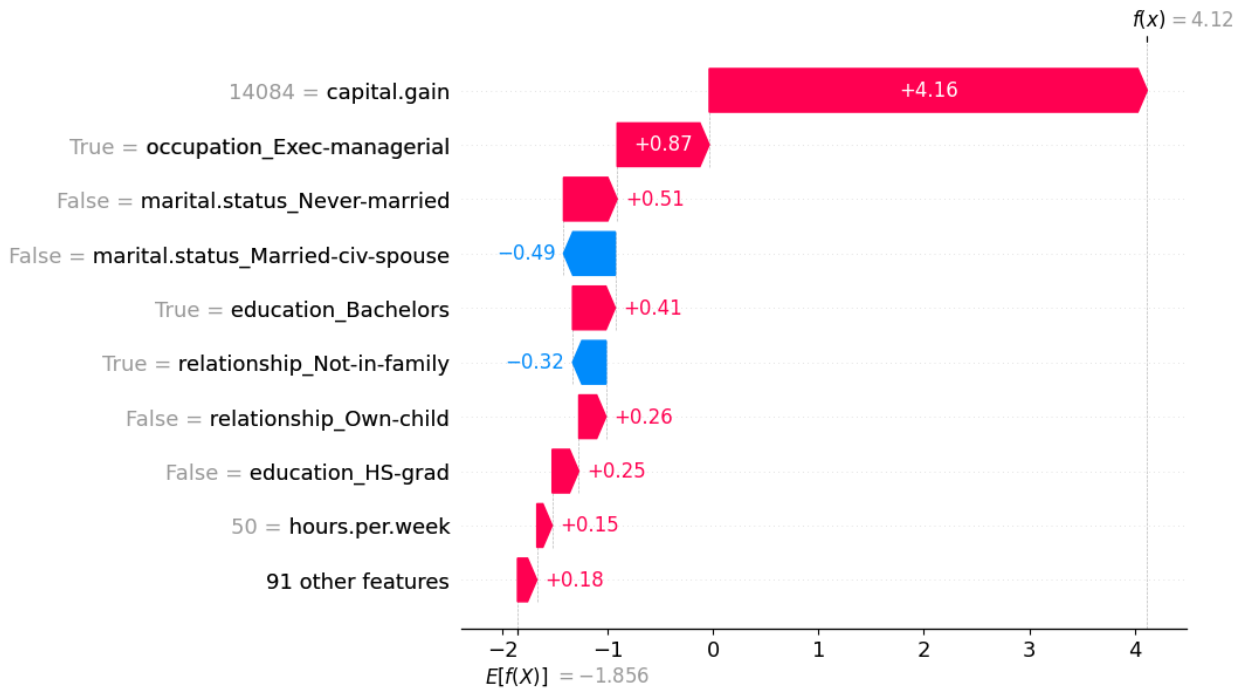
5.1 Global Explainability (SHAP)

SHAP summary plots were used to identify the most influential features in model predictions.

Key Insights:

- **Education level** was one of the strongest positive predictors of high income.
- **Hours worked per week** also significantly influenced predictions.
- **Marital status** and **occupation** contributed to income classification.

SHAP provided a global view of how the model behaves across the dataset, highlighting which features drive predictions most strongly.



5.2 Local Explainability (SHAP Waterfall Plot)

A SHAP waterfall plot was used to explain individual predictions.

Example Insight:

For a specific individual prediction:

- High education and longer working hours increased predicted income.
- Certain demographic features reduced prediction confidence.

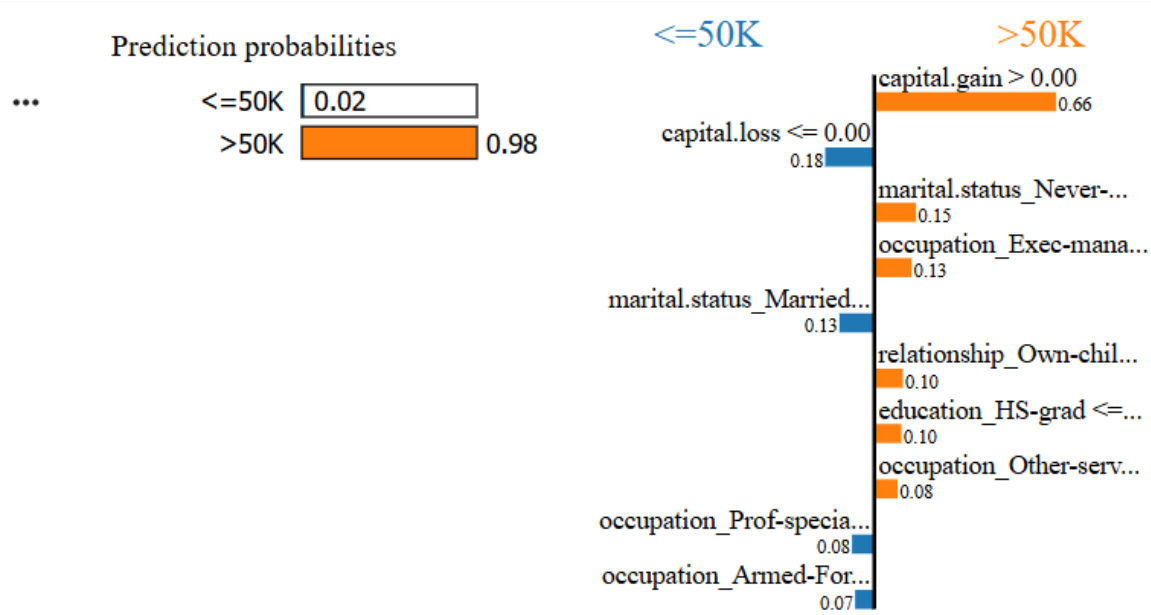
This helps explain why a particular decision was made for one individual.

5.3 Local Explainability (LIME)

LIME was applied to interpret individual predictions in a model-agnostic way.

Findings:

- LIME identified key features influencing a single prediction.
- The explanation was consistent with SHAP results but provided a simpler, local approximation.



Feature	Value
capital.gain	14084.00
capital.loss	0.00
marital.status_Never-married	0.00
occupation_Exec-managerial	1.00
marital.status_Married-civ-spouse	0.00
relationship_Own-child	0.00
education_HS-grad	0.00
occupation_Other-service	0.00
occupation_Prof-specialty	0.00
occupation_Armed-Forces	0.00

6. Ethical Considerations and Recommendations

Ethical Issues Identified:

- The model shows evidence of **gender-based disparities** in prediction outcomes.
- Bias arises from correlations in historical data rather than explicit gender usage.
- Unequal error rates may lead to unfair real-world consequences.

Recommendations for Improvement:

1. **Bias Mitigation Techniques**
    - Use Fairlearn's Exponentiated Gradient or Grid Search methods.
2. **Feature Review**
    - Remove or reduce proxy variables correlated with gender.
3. **Rebalancing Data**
    - Ensure equal representation across demographic groups.
4. **Fairness Constraints**
    - Incorporate fairness constraints during model training.
5. **Continuous Monitoring**
    - Evaluate fairness metrics before deployment in real-world systems.

7. Conclusion

This project demonstrates that while logistic regression achieves strong predictive performance on the Adult Income dataset, it may still produce **unfair outcomes across sensitive groups**.

Fairness analysis using Fairlearn revealed disparities in selection rates and error rates between gender groups. Additionally, SHAP and LIME provided valuable insights into both global and local model behavior, improving transparency and interpretability.

The findings highlight the importance of integrating fairness and explainability techniques into machine learning workflows. Future improvements should focus on bias mitigation strategies to ensure ethical and responsible AI deployment.